

Software Test Plan (Validation Specification v1)

Project: Railway Reservation System (RRS) — Team 5, UE24CS341A

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for RRS v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing the four functional modules and their supporting NFR/security requirements.

Scope: Testing covers registration/login/RBAC, inventory management, graph-based search with fare/eco-scoring, and booking/waitlist/cancellation. Excludes real payment settlement, GPS tracking, and native mobile apps (out of scope per SRS §2.3).

References: RRS SRS v1.1, RRS Software Architecture & Design Specification v1.0, PES Jackfruit – Software Engineering Project Guidelines 2026.

Definitions: PNR, RBAC, TUI, Waitlist, and Eco-score (see SRS Appendix A).

2. Test Items

- Auth & RBAC module
- Inventory & Schedule module (Admin)
- Search Engine, Fare & Eco-Optimization module
- Booking & Queue Management module
- Mock Payment interface
- ncurses TUI

3. Features to be Tested

Features mapped to SRS requirement IDs:

Requirement ID	Feature
FR-001 – FR-006	Registration, login, session, RBAC
FR-007 – FR-011	Train/route/capacity management
FR-012 – FR-017	Route search, fare, and eco-score
FR-018 – FR-024	Booking, PNR, waitlist, cancellation
NFR-001 – NFR-009	Performance, availability, usability, reliability, maintainability, testability, scalability, integrity
SEC-001 – SEC-006	Password storage, RBAC enforcement, input validation, session expiry, payment data handling, audit logging

4. Features Not to be Tested

- Real-money payment settlement / bank integration (mocked, out of scope)
- Live GPS train tracking
- Native mobile applications (only TUI and optional web wrapper are in scope)
- Multi-tenant deployment across multiple railway operators

5. Test Approach / Strategy

Levels

- Unit tests (GTest/Unity, module-level; each owner tests their own module)
- Integration tests (Search → Booking → Waitlist; Auth → all modules via RBAC)
- System tests (end-to-end TUI walkthrough)
- Acceptance tests (instructor/TA walkthrough of the demo scenarios)

Types

- Functional testing of all FR-IDs
- Negative/boundary testing (invalid input, capacity conflicts, payment failure)
- Performance testing (NFR-001, NFR-002 response-time targets)
- Static analysis and memory testing (cppcheck, Valgrind)

Entry Criteria: stable build on main, seed data (stations/trains) loaded, CI green.

Exit Criteria: all high-priority test cases executed, 0 critical defects open, $\geq 70\%$ unit coverage per module, and NFR-001/NFR-002 targets met.

5.1 Security Validation

Security objectives: SO-1 credential/data confidentiality; SO-2 restrict privileged actions; SO-3 booking integrity; SO-4 auditability.

ID	Objective	Test approach
SEC-001	SO-1	Inspect the stored user file; confirm salted hashes only, and salts differ per user.
SEC-002	SO-2	Passenger session calls an admin-only function; expect ERR_FORBIDDEN.
SEC-003	SO-3	Fuzz all input fields with overlong strings, special characters, and negative numbers.
SEC-004	SO-2	Reuse an expired/invalid session token; expect ERR_SESSION_EXPIRED.
SEC-005	SO-1	Inspect booking records; confirm no payment details are persisted.
SEC-006	SO-4	Trigger 3 failed logins and an admin action; confirm audit-log entries.

6. Test Environment

Hardware: standard developer laptops (Linux/WSL) for build and unit testing; a single Ubuntu evaluation machine for the demo.

Software: GCC/G++, ncurses, Make/CMake, gtest/Unity, cppcheck, Valgrind, Docker.

Tools: GitHub Actions (CI), GitHub Projects (backlog/board), gcov/lcov (coverage).

Test Data: synthetic seeded stations, trains, and passenger accounts (no real personal or payment data, per SRS §10).

7. Test Schedule

Milestone	Date
Test case design	02-Oct-2026
Unit tests for Sprint 1–2 modules	11-Oct-2026
Integration test execution starts.	12-Oct-2026
System / security / performance testing	19–25 Oct 2026
Final validation report	25-Oct-2026

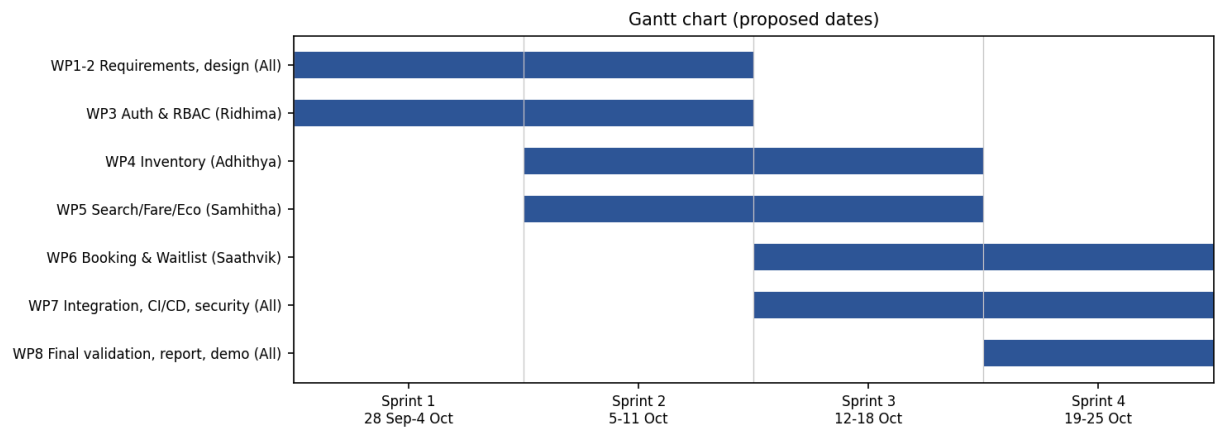


Figure 1: Sprint schedule (Gantt) across the four-week plan

8. Test Deliverables

- Test Plan (this document)
- Test cases
- Unit test report and coverage report
- CI/CD pipeline evidence
- Security validation evidence
- Test execution logs and defect reports

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Samhitha S	Prepare a plan, coordinate execution, and own search module tests.
Test Engineer	Ridhima Jain	Auth/RBAC test design & execution
Test Engineer	S S Adhithya Sriram	Inventory module test design & execution
Test Engineer	Saathvik Shenoy Padubidri	Booking/waitlist test design & execution

Product Owner / Evaluator	Course Instructor	Approve test results, sign off on readiness
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10. Risks and Mitigation

Risk	Mitigation
ncurses unavailable on evaluator's machine	Plain CLI fallback build target
Sprint 3 schedule slip (booking/waitlist complexity)	Start data structures in Sprint 2, integrate early.
Test environment inconsistency across four laptops	Standardize on a shared Docker dev image.

11. Assumptions & Dependencies

- Train, station, and route master data will be seeded by the team, not sourced from a live external database
- ncurses (or an equivalent) is available on the evaluation machine; fallback is the API + web wrapper
- GitHub Actions (or an equivalent CI tool) remains available throughout the project

12. Suspension & Resumption Criteria

Suspend testing if the build breaks on main or seed data is corrupted such that more than 30% of test cases are blocked.

Resume testing when the blocking defect is fixed and merged and the environment is verified stable.

13. Test Case Management & Traceability

The Requirement Traceability Matrix (RTM) maps SRS requirement IDs to test case IDs, extending SRS Appendix C with a System Test ID column. Example:

Req ID	Architecture ref	Design ref	Code/file ref	Test Case ID	System Test ID
FR-001	Auth & RBAC	Class: User	auth_service.*	UT-01	ST-AUTH-01
FR-012	Search/Fare/Eco	Sequence: Search Trains	search_service.*	IT-01	ST-SEARCH-01
FR-018	Booking & Waitlist	Sequence: Book Ticket	booking_service.*	IT-03	ST-BOOK-01
SEC-002	Facade / RBAC	RBAC design	middleware/auth.*	ST-01	ST-SEC-02

13.1 Detailed Test Cases

Test Case ID	Module	Type	Description	Precondition	Steps	Test Data	Expected Result	Req ID
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UT-01	Search/Fare	Unit	Fare calculation for a direct route, no concession	Graph seeded with station distances	Call fare_calculate()	500 km, Sleeper, no concession	Matches hand-computed fare	FR-014
UT-02	Search/Fare	Unit	Fare calculation with senior concession	Graph seeded	Call fare_calculate() with concession flag	500 km, Sleeper, senior	Discounted fare returned	FR-014
UT-03	Search	Unit	Reject invalid station code	Graph loaded	Search with a bad station code	Origin='XXX'	ERR_INVALID_INPUT	FR-012
IT-01	Search	Integration	Direct route search returns all matching trains	Seeded network A-B with 3 trains	search_routes('A','B',date)	A → B	All 3 direct trains listed, ranked	FR-012
IT-02	Search	Integration	Connected (multi-leg) route search when no direct train exists	No direct A-C train; B connects them	search_routes('A','C',date)	A → C	Multi-leg route via B returned	FR-013
IT-03	Booking	Integration	Booking is not created when mock payment fails	Valid search result selected	booking_create() with payment mocked to fail	Payment=FAIL	No PNR created; ERR_PAYMENT_FAILED	FR-019
IT-04	Booking	Integration	Waitlist enqueue and automatic promotion on cancellation	Train/class at full capacity	Book 1 more (waitlisted), then cancel 1 confirmed booking	Capacity=N, booking_s=N+1	New booking WAITLISTED, then promoted to CONFIRMED after cancellation	FR-020, FR-022
ST-01	Auth/RBAC	System	Passenger session cannot call an admin-only function	Logged in as passenger	Call train_create() with passenger session	role=PASSENGER	ERR_FORBIDDEN returned	FR-004, SEC-002
ST-02	Search	System / Performance	Search responds within the NFR-001 time budget	200 stations, 500 trains seeded	Run 100 concurrent search queries	Random station pairs	≥95% of queries return within 2 seconds	NFR-001

ST-03	Booking	System	Duplicate booking is not created on payment-timeout retry	Booking request sent once, times out	Retry the identical booking request	Same idempotency key	Exactly one booking exists	NFR-005, FR-024
ST-04	Auth	Security	Passwords are never stored in plaintext	User registered	Inspect the persisted user record	N/A	Only a salted hash is present; salts differ per user	SEC-001
ST-05	Booking	Security	No seat is double-booked under concurrent requests	Class has exactly 1 seat left	Fire 50 concurrent booking requests for that seat	Concurrent load=50	Exactly 1 CONFIRMED booking; rest WAITLISTED	NFR-009

14. Test Metrics & Reporting

- % of test cases executed and % passed/failed
- Defect density and defect aging
- Requirement coverage (RTM completeness)
- Unit test line/branch coverage per module

Reports: sprint-end execution status, final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA Lead	Samhitha S	
Dev Lead		
Course Coordinator		